

A Lightweight Hybrid Rule-Based Framework for Interpretable Phishing Detection

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Abstract—The phishing attacks exploit the dependency on the URL deception, which is a consistent cybersecurity risk. The current techniques of attack recognition based on blacklists and machine learning are ineffective against new types of phishing attempts, require heavy computational capacity, and do not ensure any transparency in recognizing the threats. In this research work, we present the method of detection of the phishing attacks based on a hybrid rules-based framework. The system under discussion utilizes blacklist checking, heuristics of URL analysis, the analysis of the similarity between domains, hyperlink detection, and linguistic analysis for comprehensive attack detection. The scoring system produces the transparency of suspiciousness scoring, ensuring prompt detection and rational decision making by users. Experimental evaluation exhibits high performance, achieving 92.53% accuracy, 99.23% precision, 85.73% recall, and 99.33% specificity, while continuing light computational overhead. These results show an effective, interpretable, and scalable solution for phishing detection without dependence on complex machine learning models.

Keywords—Phishing Detection, URL Analysis, Heuristic-Based Detection, Lightweight Security Framework, Rule-Based System.

I. INTRODUCTION

The rapid development of information technology systems along with the growing use of online services has led to new challenges in information security. In particular, users face significant threats, including monetary and personal losses [1]. In this regard, phishing stands out among the risks as a social engineering attack where criminals take advantage of users' vulnerability to obtain sensitive information. Criminals often masquerade as trusted companies by sending fake emails, establishing malicious sites, and using deceptive domain names to gain illegal access to the victims' personal information and accounts [2].

Phishing remains a persistent threat worldwide. According to the report issued by the Anti-Phishing Working Group (APWG), there were about 892,494 cases of phishing attacks in Q3 2025. SaaS/webmail was responsible for 21.2% of phishing attacks, whereas social networks made up 14.6%. Other sectors targeted include financial organizations. As seen from the statistics, phishing attacks mainly target popular resources; thus, many users can potentially fall victim to these attacks [3].

A frequent method for phishing attacks involves malicious URLs that redirect users to fraudulent websites. Attackers commonly exploit trusted platforms and link aggregation services to improve the credibility of these links, taking advantage of their ease of access and minimal verification requirements [2]. Beyond URL-based attacks, phishing techniques have evolved to include QR codes as a delivery vector. Malicious QR codes, typically distributed via emails or attachments, can direct users to harmful websites or initiate unintended actions upon scanning [4]. These threats are notably concerning because QR codes are not easily analyzed by standard filtering systems. Recent reports identified 716,306 unique malicious QR codes in the third quarter of 2025, showing a 13% rise from the previous quarter and surpassing 3 million over a 12-month period [3].

Numerous anti-phishing techniques have been developed to counter phishing attacks. These methods are commonly categorized as list-based approaches, heuristic procedures, mixed approaches, and information flow-based techniques. Blacklist systems, a type of list-based approach, allow for rapid detection yet are limited to previously identified threats, making them ineffective against newly generated or obfuscated phishing URLs [5]. The heuristics procedure exploits the structural and linguistic elements of URLs in detecting malicious characteristics, thereby providing enhanced adaptability. Nevertheless, designing such rules requires precision. The machine learning and deep learning techniques provide superior detection rates. However, their reliance on huge datasets, complicated feature extraction process, and continuous relearning makes them expensive processes unsuitable for real-time detection [6], [4].

As a means to overcome these drawbacks, this paper proposes the development of a lightweight and training-free phishing detector using a hybrid architecture. The proposed scheme uses a multi-faceted identification approach which combines the different complementary methods in order to enable comprehensive analysis of various forms of phishing attacks involving URLs and QR codes [7], [8]. The integration of the methods, including blacklists check, heuristics, domain similarity, and hyperlink filters allows for accurate real-time detection without the need for huge datasets and complicated machine learning processes [1], [4],

[9]. Furthermore, the system has a scoring mechanism whereby it provides a percentage figure indicating the extent of suspicion of a particular URL input by the user [10], [11].

II. RELATED WORK

This chapter reviews the basic approaches and the previous research done in phishing detection, with special emphasis being placed on the approaches that facilitate fast, understandable and real-time processing.

A. Blacklist

Methods using blacklists detect and prevent phishing attempts through the comparison of URLs against known lists of malware websites [13]. These lists are compiled from user reports, automatic systems, and threat intelligence feeds, and they can be searched either locally or remotely. Such an approach is quick, efficient, and requires little computing power, making it suitable for live systems [8].

The main problem with blacklist approaches is that they will only prevent phishing attacks on known URLs but not those on new ones. As phishing is continually evolving, many potentially harmful URLs slip through, despite updates to the blacklists on a regular basis [5]. The reliance on user reports and manual additions results in delays, thus prolonging the operational time of phishing websites [13].

B. Heuristic

The use of heuristics in detecting phishing websites eliminates the weaknesses of using blacklist detection since they rely on examining the structure and language used by the URL rather than relying entirely on static databases [5], [8]. In the case of using heuristics in detecting phishing websites, patterns of phishing are examined, including IP-based URLs, domain structure, and unusual symbols [10], [14].

Despite their effectiveness in detecting new phishing sites, these approaches have a number of shortcomings. For instance, attackers can copy the structure of real domains and also use other complex methods, thereby making it difficult for such methods to detect phishing sites [11]. Recent methods used in detecting phishing have involved combining heuristic features with other approaches like content analysis and string similarity analysis (Longest Common Subsequence and Edit Distance) [1], [15].

C. Domain Similarity

In This Paper, Domain Similarity is used to identify any phishing cases where there is domain squatting or URL manipulation. In this paper, malicious domain properties (D1D5) are obtained by analyzing the degree of similarity between the domain/subdomain under test and a set of brand domains that have been previously identified as high-risk. The approximate string matching algorithm in the fuzz library allows the discovery of brand names even if they are not present as a whole but only as part of the string, or even when there are errors in spelling. If the degree of similarity exceeds the threshold value, then the domain is branded as a phishing site [10].

D. Hyperlink and Lexical Analysis

The paper points out some attributes of the URL and hyperlink which are useful in spotting phishing sites. One of the features highlighted is the use of an IP-based URL which sees attackers utilizing an IP address instead of domain names, thus bypassing domain name registration and DNS; in this case, the feature will be given a weight of 1 while those without will receive a score of 0. Age of domain name is

another important feature, especially in detecting phishing sites, as most such sites do not have an extended lifespan. In this regard, any domain name with a year of age or younger will get a weight of 1 after WHOIS look up. Other attributes include the length of URL, as phishing sites usually make their URLs overly lengthy in an effort to conceal malicious elements; URLs longer than 53 characters will get a score of 1. The dots found within the hostname will also play an important part in identifying phishing attacks; those having more than three dots are deemed suspicious [1].

Furthermore, this paper highlights the lexical features that are based on the text-related characteristics of URLs for detecting phishing websites. The important lexical features considered for this purpose include URL length, the digit ratio (ratio of digits in the URL), presence of questionable symbols like @, #, and \$ in URLs, and URL entropy that represents the degree of randomness of characters in URLs. In addition, other structural characteristics like subdomains, directory depth, and query parameters can be added to this feature set.

III. METHODOLOGY

In this research, attention is drawn to creating a system that detects phishing attacks by identifying suspicious link properties and computing a numeric risk score. The primary innovation lies in the design of the rule-based detection algorithm along with a corresponding risk scoring technique for classification.

A. Research Workflow

The proposed framework is the workflow Fig. 1

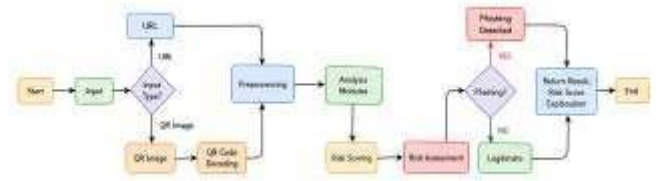


Fig. 1. Research Workflow

B. System Architecture

The presented phishing detection architecture is designed as a backend application for live data processing without the need for training a model or using any dataset [7]. The backend architecture is implemented using FastAPI, hosted with the help of the Uvicorn web server, and provided through a RESTful interface that is described using the OpenAPI 3.1 specification. The API is designed to accept URL, decoded QR code content, and QR images, where the latter is decoded to extract an embedded URL [9]. The provided APIs include endpoints to process URLs, QR code content, QR images, and provide monitoring capabilities with a structure output including the result of the analysis and classification. Input URLs are preprocessed and analyzed within a multilayer pipeline consisting of blacklist, heuristics, and similarity-based approaches with hyperlinks and lexicons used accordingly [1]. Output results are aggregated in a risk score layer for calculation of a numerical value and a percentage, along with mapping internal classification results into two categories of binary user-friendly outputs.



Fig. 2. Architecture of the Phishing Detection System

C. Data Collection

This research requires the use of external reference data for the development of a no-training, rule-based phishing detection framework [7]. These data will be used to develop detection rules, perform blacklisting verification, analyze domain similarities, and evaluate the developed system. In this respect, phishing URLs are obtained from PhishTank and OpenPhish in amounts of 57,212 and 300, respectively [13], [16]. The matching process between the provided data uses exact URL and host-path matching methods to increase accuracy and minimize false positives. The proposed methodology includes the development of 25 heuristic rules, considering the following features: excessive URL length, presence of suspicious characters, unusual domain structure, and misleading keywords [5], [11]. In addition, legitimate domains are retrieved from trusted resources such as Tranco, Majestic, and high-priority brand sites to perform the analysis of lookalike domains and typo-squatting attacks [17]. Moreover, hyperlink and lexical analysis rules are created according to the URL structure and hyperlinks used within the URLs [10]. As an evaluation measure, a well-balanced direct-URL dataset is employed, which consists of 750 phishing URLs and 750 legitimate URLs [18].

D. Detection Mechanism

The new technique for detecting phishers incorporates various complementary modules in order to eliminate the drawbacks associated with detection using single techniques [1], [10]. The major components for analysis include blacklist, heuristic, hyperlink, lexical, and similarity analysis. However, all the modules are not equally effective because blacklist, heuristic, and similarity analysis make up the primary pipeline, whereas the remaining two are auxiliary layers of webpage evidence [5], [14], [16].

1) Blacklist Verification

For blacklists verification, in the current paper, a hierarchical matching mechanism using normalized URL input has been introduced as the first detection layer in the system. The detection starts from exact URL matching. If no such match is achieved, host-path matching follows. Both approaches are considered strong phish detection mechanisms that have a direct impact on final classification decisions. However, host-only matching has been omitted in order to prevent false positives for legitimate services running on shared servers [9]. This allows not only preserving blacklist detection but also making it possible to implement further

layers such as heuristic and similarity analysis [1]. The described blacklist workflow is presented in Fig. 3 below.

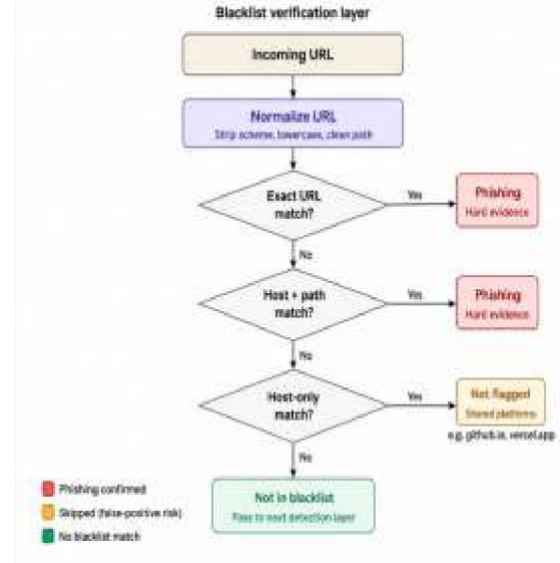


Fig. 3. Blacklist Workflow

2) Heuristic URL Analysis

In the proposed system, heuristic analysis is adopted for analyzing structural and lexical features of the URLs involved in the detection process [5], [8]. The system analyzes parameters such as suspicious top-level domains, IP addresses in URLs, use of punycode, long URLs, several subdomains, multiple hyphens, path strings for login actions, suspicious file path strings, and brand-name obfuscation strings [10], [14]. In contrast to considering individual parameters for detection, emphasis is placed on combinations of these features in order to enhance the accuracy of detection while minimizing false positives [1], [11]. The heuristic module employs an additive score method, whereby each feature increases the risk factor score from 0.0 to 1.0.

3) Hyperlink Analysis

Hyperlink analysis serves as supplementary contentbased evidence in the presence of webpage contents [1], [10]. It evaluates webpage forms and interactive components, such as password entry fields, authentication forms, external forms submissions, suspicious redirections, inconsistent titles, and verifications-related terminology [2], [11]. While not designed to serve as the main phishing detection technique, hyperlink analysis enhances reliability when the URL lacks sufficient evidence [19]. It comes in handy when dealing with phishing webpages that reside within legitimate infrastructures since their behavioral characteristics indicate phishing attacks [16]. Fig. 4 provides a visualization of the proposed Hyperlink flow.

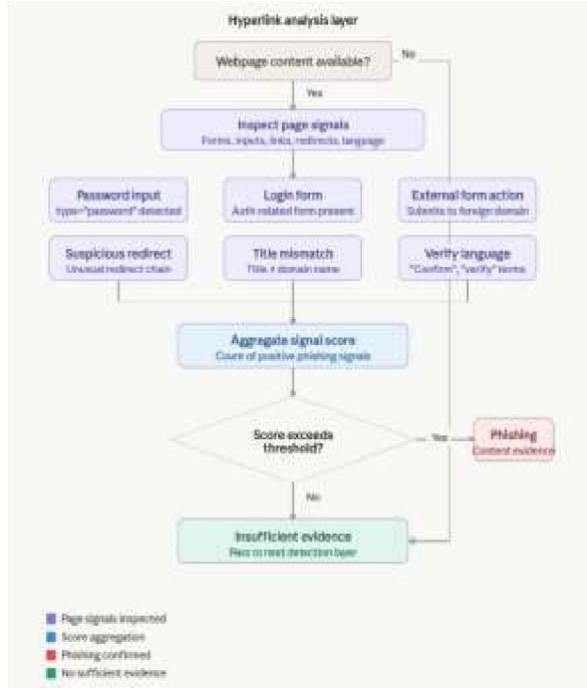


Fig. 4. Hyperlink Workflow

4) Lexical Analysis

Lexical Analysis focuses on analyzing the webpage text, including the title of the webpage and its content, to find words or phrases associated with phishing attacks or contradictions between the content of the webpage and the website's domain name [5], [14]. The Lexical Analysis component can recognize the presence of words that relate to brands, words that usually occur in phishing attacks, such as login, verify, password, confirm, and account, as well as conditions where there are brand-domain conflicts [10], [18]. For this research, lexical analysis is considered as evidence, not as a sole basis for deciding whether an attack is taking place [8].

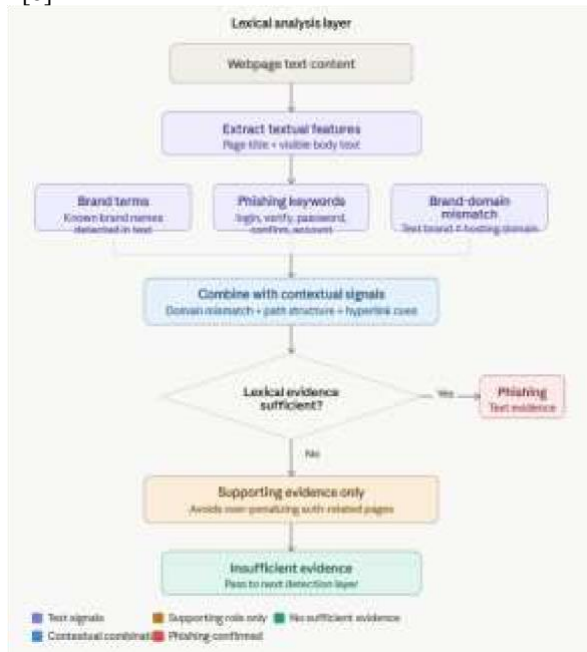


Fig. 5. Lexical Workflow

5) Similarity Analysis

A similarity method is implemented for the detection of phishing attacks via look-alike domains and typo-squatting by mimicking trusted brands [17], [20]. Within the designed framework, a similarity process is performed between the registered domain name of a given URL and a list of known legitimate domains. The similarity is determined using the Sequence Matcher and edit distance technique in order to recognize substituted characters, typos brand names, and domain masquerading [15], [21]. While there will be no penalty for exact matching, any look-alike domains will incur a high similarity risk score [10].

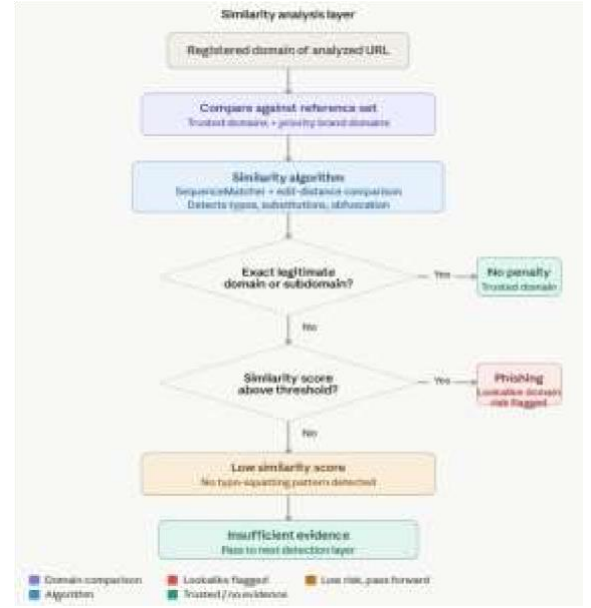


Fig. 6. Similarity Workflow

E. Scoring System

The final result is obtained through the use of a riskscoring framework, which merges the results from both detection modules into one phishing risk score [1], [12]. The risk scoring system is deliberately kept simple for easy interpretation and understanding of the contribution of each individual signal.

The base score is given by (1):

$$Score_{base} = (wH \times H) + (wB \times B) + (wS \times S) \quad (1)$$

Where H is the heuristic value, B is the blacklisted value, S is the similarity value, and wH, wB, and wS are the weighting factors. In the present implementation, the values of the weights are set to wH = 0.35 and wB = 0.20, and the heuristic value is further subdivided as follows (see Eq. 2):

$$H = (0.75H_{url}) + (0.15H_{link}) + (0.10H_{lex}) \quad (2)$$

Whereas H_{link} represents heuristic evidence based on hyperlinks and H_{lex} represents lexical evidence, H_{url} refers to heuristic evidence based on URLs. Following the computation of the base score, the system awards additional bonuses and deductions to indicate phishing or safe combinations, as illustrated in (3):

$$Score_{final} = Score_{base} + Bonus + Penalty \quad (3)$$

This result is then truncated to [0,1], and the risk percentage is calculated according to (4):

The internal workings involve three different analytical classes:

- $Score_{final} < 0.34$ indicates legitimate,
- $0.34 \times Score_{final} < 0.60$ indicates suspicious
- $Score_{final} \times 0.60$ indicates phishing (4)

For user-oriented classification purposes, legitimate URLs are categorized as non-phishing URLs, whereas suspicious and phishing URLs are both categorized as phishing URLs. The reason behind this is that it helps keep the binary output of the model intact while providing more information for internal analysis purposes [11]. When it comes to evaluating the performance of the model, the confusion matrix is used, which involves accuracy, precision, recall, specificity, F1-score, balanced accuracy, and Matthews correlation coefficient (MCC) [18], [19].

F. Evaluation

The system performance was assessed based on the confusion matrix metrics such as accuracy, precision, recall, specificity, F1-score, balanced accuracy, and Matthews correlation coefficient (MCC) [18], [19]. Such criteria were chosen due to the insufficiency of accuracy alone to evaluate a binary classifier sufficiently and properly, where various errors differ greatly from each other in meaning [11]. The results of confusion matrices were achieved in absolute numbers and percentages of all examined URLs.

Accuracy reflects the overall share of the correctly predicted results among the entire set. Precision means the percentage of truly phishing URLs predicted as phishing. Recall means the number of truly phishing URLs that were detected by the algorithm. Specificity shows the amount of legitimate URLs which were properly recognized as safe. F1score shows the balance of precision and recall. Balanced accuracy shows the average accuracy in predicting phishing and legitimate classes. MCC shows the stability of the result by considering all elements of the confusion matrix [18].

IV. EXPERIMENTAL RESULTS

A test dataset consisting of balanced phishing URLs and normal web addresses was used to determine the performance of the developed classifier under real-world scenarios. Being rule-based, the classifier performs well when assessed on a basis of binary classification based on confusion matrices. The performance criteria of the proposed phishing URL detection classifier have been described in Fig. 7

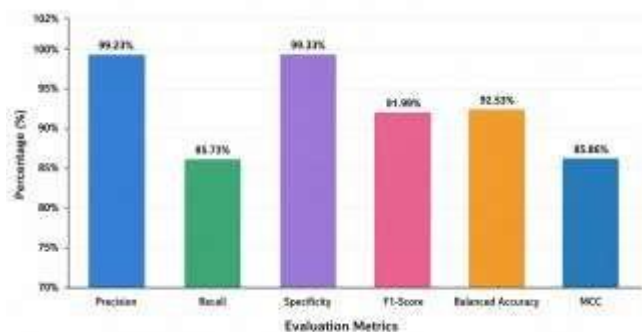


Fig. 7. Evaluation Metrics

At the stage of evaluating the model, the outcome was considered to be a binary classification, either non-phishing or phishing. Within the model, there are legitimate, suspicious, and phishing categories maintained; however, suspicious and phishing classifications are merged in the decision-making process into a single category phishing. Confusion matrix for this is provided in Table I.

Table I
Confusion Matrix

Metrics	True Positive	True Negative	False Positive	False Negative	Total
Total Number	643	745	5	107	1500
Percent age	42.87%	49.67%	0.33%	7.13%	100%

From the above experimental results, it can be concluded that the suggested approach is efficient for the detection of phishing URLs. High precision (99.23%) and high specificity (99.33%) suggest that the framework is capable of producing minimum false alarms. This is mainly because of the blacklist validation component, which makes accurate predictions using strict URL and host-path matching as well as conservative heuristics for preventing unnecessary flagging of legitimate websites. The attained recall rate of 85.73% shows that the system successfully detects a considerable number of phishing URLs, which is mainly attributed to the role of heuristics and domain name comparison techniques. These factors help the system to detect novel phishing URLs that are not available in the blacklist, such as lookalike domains and unusual lexicon structures. On the other hand, the recall rate implies that some advanced or less visible phishing URLs could be left undetected.

F1-score (91.99%) and Balanced Accuracy (92.53%) are an example of a very good balance between precision and recall; hence, the use of several detection layers works very well. Specifically, blacklist verification is good in terms of precision, but at the same time, heuristic and similarity methods offer better recall. MCC is an indicator of very good correlation between the predicted and real data, showing that the proposed method performs excellently on both phishing and non-phishing domains.

Overall, the proposed framework worked effectively by integrating blacklist verification, heuristic analysis, and domain similarity techniques. During the evaluation of a balanced dataset consisting of 500 URLs, the framework maintained an average CPU utilization of 9.42% and an average memory usage of 183.11 MB, with an average response time of 4944.24 ms (about 10 seconds) per URL and a throughput of 0.202 URLs/s, demonstrating low computational resource requirements throughout the evaluation.

V. CONCLUSION

The proposed system is designed to be an efficient and lightweight system that would allow for analyzing either direct links or QR code-based inputs. Multiple phishing

detection techniques were incorporated in the developed system and include blacklisting, heuristic detection, hyperlink checking, lexical checking, and domain name checking; in addition, the risk scoring mechanism was used in the process to achieve high levels of interpretability. According to the findings of the experiment, the system performed very well, having achieved 92.53% accuracy, 99.23% precision, and 99.33% specificity, which proves high effectiveness in detecting phishing sites, as well as the ability to do it without producing too many false-positive predictions. At the same time, some limitations exist in the ability of the system to detect phishing sites; in particular, such types of attacks as zero-day attacks or new ones may not be detected because of the use of predefined rules. However, in order to avoid risks associated with them, the risk score can help evaluate the threat. Compared with typical machine learning-based phishing detection systems, which generally require model loading and higher computational resources, the proposed framework is designed to operate with minimal resource requirements.

Author Contributorship

Muhammad Adam Zuhdi: Conceptualization, Methodology, Data Curation, Formal Analysis, Investigation, Visualization, Writing – Original Draft Preparation. **Maghfirli Al Ayubi:** Software, Data Curation, Formal Analysis, Validation, Writing – Review and Editing. **Maria Rianti Gadi Djou:** Methodology, Validation, Visualization, Formal Analysis, Writing – Review and Editing. **Ayu Maulina:** Supervision, Validation, Review.

Data Availability

We demonstrate our commitment to data openness and transparency. To facilitate further research, we have made the data used in their study publicly available.

The data used by the author can be opened via the link below:

- <https://github.com/PHISHING-DETECTIONSKRIPSI/main.git>

REFERENCES

- [1] G. Sonowal and K. S. Kuppasamy, "PhiDMA – A phishing detection model with multi-filter approach," *Journal of King Saud University - Computer and Information Sciences*, vol. 32, no. 1, pp. 99–112, Jan. 2020, doi: 10.1016/j.jksuci.2017.07.005.
- [2] R. Alabdan, "Phishing attacks survey: Types, vectors, and technical approaches," Oct. 01, 2020, MDPI AG. doi: 10.3390/fi12100168.
- [3] Anti-Phishing Working Group, "Phishing Activity Trends Report, 3rd Quarter 2025," 2025.
- [4] A. S. Rafsanjani, N. B. Kamaruddin, H. M. Rusli, and M. Dabbagh, "QsecR: Secure QR Code Scanner According to a Novel Malicious URL Detection Framework," *IEEE Access*, vol. 11, pp. 92523–92539, 2023, doi: 10.1109/ACCESS.2023.3291811.
- [5] R. Sakunthala Jenni and S. Shankar, "Review of various methods for phishing detection," *EAI Endorsed Transactions on Energy Web*, vol. 18, no. 20, 2018, doi: 10.4108/eai.12-9-2018.155746.
- [6] D. Forensics and A. Applications, "Phishing URL Detection: a Comparative Study of Heuristic Methods, Machine Learning, and Temporal Analysis(hal:26)."
- [7] A. Butnaru, A. Mylonas, and N. Pitropakis, "Towards lightweight urlbased phishing detection," *Future Internet*, vol. 13, no. 6, Jun. 2021, doi: 10.3390/fi13060154.
- [8] C. M. R. da Silva, E. L. Feitosa, and V. C. Garcia, "Heuristic-based strategy for Phishing prediction: A survey of URL-based approach," *Comput. Secur.*, vol. 88, p. 101613, Jan. 2020, doi: 10.1016/j.cose.2019.101613.
- [9] G. A. Amoah and H.-A. J.B., "QR Code Security: Mitigating the Issue of Quishing (QR Code Phishing)," *Int. J. Comput. Appl.*, vol. 184, no. 33, pp. 34–39, Oct. 2022, doi: 10.5120/ijca2022922425.
- [10] R. Goenka, M. Chawla, and N. Tiwari, "Enhanced Phishing Detection Approach Using a Layered Model: Domain Squatting and URL Obfuscation Identification and Lexical Feature-Based Classification," *IEEE Access*, vol. 13, pp. 187285–187306, Oct. 2025, doi: 10.1109/ACCESS.2025.3626819.
- [11] S. Kailas and R. Roopalakshmi, "'Think Before You Click'—Malicious URL Detection in Cybersecurity: A Systematic Review and Research Roadmap," *IEEE Access*, vol. 13, pp. 154305–154325, 2025, doi: 10.1109/ACCESS.2025.3601387.
- [12] D. Duah, B. K. O. Sarfo, C. Boakye, and D. Duku, "PHISH OR LEGIT? QUANTIFYING WEB-BASED THREAT SIGNALS THROUGH PREDICTIVE ANALYTICS AND FEATURE ATTRIBUTION," *International Journal of Professional Business Review*, vol. 10, no. 5, p. e05472, May 2025, doi: 10.26668/businessreview/2025.v10i5.5472. [13] S. Bell and P. Komisarczuk, "An Analysis of Phishing Blacklists: Google Safe Browsing, OpenPhish, and PhishTank," in *ACM International Conference Proceeding Series, Association for Computing Machinery*, Feb. 2020. doi: 10.1145/3373017.3373020.
- [14] M. Tellakula, P. Kanduri, S. R. Ramireddy, and U. R. Konche, "Machine Learning-Based Phishing URL Detection Using Lexical and Structural Features," *Int. J. Comput. Appl.*, vol. 187, no. 96, pp. 1–5, Apr. 2026, doi: 10.5120/ijcaeeeb7bb333.
- [15] J. Nikonowicz, L. Matuszewski, and P. Kubczak, "Sequence Alignment Algorithm for Statistical Similarity Assessment," *IEEE Access*, vol. 9, pp. 102153–102160, 2021, doi: 10.1109/ACCESS.2021.3098340.
- [16] A. Oest et al., {PhishTime}: Continuous Longitudinal Measurement of the Effectiveness of Anti-phishing Blacklists. 2020. Accessed: Apr. 18, 2026. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity20/presentation/zhou-jie>
- [17] A. Prasad and S. Chandra, "PhiUSIIL: A diverse security profile empowered phishing URL detection framework based on similarity index and incremental learning," *Comput. Secur.*, vol. 136, p. 103545, Jan. 2024, doi: 10.1016/J.COSE.2023.103545.
- [18] A. Karim, M. Shahroz, K. Mustofa, S. B. Belhaouari, and S. R. K. Joga, "Phishing Detection System Through Hybrid Machine Learning Based on URL," *IEEE Access*, vol. 11, pp. 36805–36822, 2023, doi: 10.1109/ACCESS.2023.3252366.
- [19] M. Sanchez-Paniagua, E. F. Fernandez, E. Alegre, W. Al-Nabki, and V. Gonzalez-Castro, "Phishing URL Detection: A Real-Case Scenario Through Login URLs," *IEEE Access*, vol. 10, pp. 42949–42960, 2022, doi: 10.1109/ACCESS.2022.3168681.
- [20] Y. Wang, W. Li, R. Xiao, J. Sun, and S. Jin, "Graph-based Malicious Domain Name Detection: How to Use the Heuristic Relations," *IEEE Trans. Dependable Secure Comput.*, 2025, doi: 10.1109/TDSC.2025.3630625.
- [21] T. P. Rinjeni, A. Indriawan, and N. A. Rakhmawati, "Matching Scientific Article Titles using Cosine Similarity and Jaccard Similarity Algorithm," *Procedia Comput. Sci.*, vol. 234, pp. 553–560, 2024, doi: 10.1016/j.procs.2024.03.039.